

FINAL EXAM PORTFOLIO · ITAI-1370

# Artificial Intelligence History, Theory & Platforms

A complete record of the Summer 2026 course journey —  
13 modules, real work, real reflection.

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TERM Summer 2026

**13/13**

MODULES COMPLETE

# Course Grade Snapshot

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**97%**

ASSIGNMENTS

**99%**

PUZZLES

**94%**

MIDTERM

**13**

MODULES  
DOCUMENTED

This portfolio is built directly from module-by-module documentation maintained throughout the semester in the course repository ([github.com/ClayClimate-AI/ITAI-1370-history-theory](https://github.com/ClayClimate-AI/ITAI-1370-history-theory)) — real submitted work, real grades, and honest reflection. Nothing in this document is invented; every module below is a synthesis of work already completed.

## TABLE OF CONTENTS

- 01** Introduction to A.I.: Course Setup
- 02** Introduction to A.I.: The Big Issues
- 03** Games, Prelude to A.I.
- 04** Games Change Everything
- 05** Machine Learning: The Data
- 06** Machine Learning: The Pipeline
- 07** Deep Learning: Neural Networks
- 08** Deep Learning: Big Data and Architectures
- 09** Computer Vision: Image Processing
- 10** NLP: Basics
- 11** NLP: LLMs
- 12** AI Agents
- 13** Robotics: Grand Tour
- Midterm Exam — AI Glossary
- ★ Bonus — Building an Agent: Hermes
- Conclusion & Future Objectives

# Module-by-Module Record

## MOD 01 Introduction to A.I.: Course Setup

**ACTIVITIES** Wrote a full research paper ("The Evolution of Artificial Intelligence") tracing AI from myth and philosophy through the Dartmouth Conference, the symbolic-AI era and AI winters, to deep learning and modern LLMs. Completed the Glossary Kick Start lab.

**WHAT I LEARNED** The "AI effect" — each generation redefines intelligence as whatever machines still can't do — explains why AI always feels both overhyped and perpetually out of reach. The deeper throughline of the whole paper: AI's history is a repeating cycle of bold ambition, disappointment, and a sidelined idea (connectionism/neural networks) eventually returning to redefine the field.

A01 — 100%

L01 — 100%

## MOD 02 Introduction to A.I.: The Big Issues

**ACTIVITIES** Researched why Google was founded (PageRank vs. keyword-frequency search) and analyzed the 2014–2023 GenAI patent surge (A02). Built an 11-person research spreadsheet tracing foundational AI figures — Turing, McCarthy, Minsky, Shannon, Rosenblatt, Hinton, LeCun, Bengio, Vaswani et al., Samuel — from their landmark work to a named system in use today (L02). Solved Google's 100-prisoner hat puzzle using parity encoding and mapped it to AI concepts: error correction, binary classification, Bayesian inference (Puzzle 02).

**WHAT I LEARNED** Nearly every foundational AI breakthrough documented in L02 was funded by DARPA, ONR, NSF, or internal corporate R&D — AI research has never been separable from defense and corporate funding. From the hat puzzle: one bit of well-chosen shared information is enough to turn a group of individually-blind agents into a system with 99% guaranteed accuracy — the rule is the intelligence, not any single agent's knowledge.

*Screenshot placeholder: hat-puzzle simulator images already saved in `modules/module-02/resources/`*

A02 — 98%

L02 — 98%

Puzzle 02 — 99/100

## MOD 03 Games, Prelude to A.I.

**ACTIVITIES** Built a client-side AI Stock News Sentiment Tracker (rule-based scoring: weighted keywords, negation detection, intensity modifiers) for the InClass assignment. Analyzed AlphaStar's limitations (narrow intelligence, lifetime adaptation, forward-model planning) for A03. Built a Scratch paddle-and-ball game for L03, debugging a real race-condition bug in the score system. Solved the Monty Hall problem using Bayes' Theorem for Puzzle 03.

**WHAT I LEARNED** Mastery inside a fixed, fully-specified environment (a video game, a puzzle with a formula) doesn't transfer to open-ended environments — true for both AlphaStar and for naive probability intuition. The Scratch score-reset bug was a small, visible version of a real coordination hazard: multiple independent triggers touching shared state.

InClass — 98/100

A03 — 98/100

L03 — 98%

Puzzle 03 — 99/100

## MOD 04 Games Change Everything

**ACTIVITIES** Researched raytracing vs. rasterization, real-time ray tracing (NVIDIA RTX), and AI-accelerated rendering/DLSS for A04. Built a visual timeline of 8 milestones in computer graphics rendering history (1960s ray casting → 2020+ AI-accelerated path tracing) using generative AI (DALL-E) imagery for L04, deployed as a live interactive site.

**WHAT I LEARNED** Raytracing's cost isn't a flaw — it's the direct tradeoff for eliminating lighting approximations; AI upscaling (DLSS) is what's currently shrinking that cost. On generative AI as a research tool: prompt quality is entirely bottlenecked by the prompter's existing subject knowledge, which limits its use as an independent educational resource.

A04 — 98%

L04 — 98%

## MOD 05 Machine Learning: The Data

**ACTIVITIES** Researched ChatGPT's three-stage training (pretraining → supervised fine-tuning → RLHF) and ran six real test questions against it, critically analyzing accuracy/human-likeness/limitations (A05). Researched three computational humor systems (JAPE, GPT-3, Bard/Gemini), designed a manual joke-generation algorithm, and tested a real joke on three human respondents (L05).

**WHAT I LEARNED** RLHF optimizes a model to *sound* aligned with human preference — not to know when it's uncertain. Those are different problems. From the humor lab: an algorithm can verify a joke's mechanical structure but can't predict whether a given audience's background will make the connection land — mechanics are computable, impact isn't.

A05 — 98%

L05 — 95%

## MOD 06 Machine Learning: The Pipeline

**ACTIVITIES** Built a knowledge graph of the ML landscape (Foundations, Learning Approaches, Techniques, Tools/Frameworks, Applications) in Whimsical, based on Daniel Bourke's ML Roadmap, explicitly mapping cross-branch dependencies.

**WHAT I LEARNED** The ML landscape isn't a sequential curriculum — a few nodes (Python, Cloud Platforms, Version Control) function as infrastructure every other branch depends on simultaneously, not a prerequisite finished once and left behind.

Image available:

`modules/module-06/assignments/machine-learning-knowledge-graph.png`

A06 — 95%

## MOD 07 Deep Learning: Neural Networks

**ACTIVITIES** Designed a feed-forward neural network predicting HVAC RTU failure risk from real field-relevant features (unit age, days since service, ambient temp, fault codes), walking through a full forward pass and backpropagation correction on a genuinely ambiguous test case (A07). Ran all 5 TensorFlow Playground experiments — activation functions, neuron count, learning rate, data noise, dataset complexity — with real loss numbers for each (L07).

**WHAT I LEARNED** A neural network doesn't need a human to write the rule "old AND recently-serviced = lower risk" — it derives that interaction from a single wrong, confident prediction and the gradient of that error. From L07: a neural network's success isn't a property of the network alone — it's the fit between architecture, hyperparameters, and data (the same 4-neuron network went from perfect on Gaussian data to nearly useless on a spiral).

A07 — 94%

L07 — 94%

**MOD 08****Deep Learning: Big Data and Architectures**

**ACTIVITIES** Designed and ran a story-based Generator/Discriminator activity to embody GAN mechanics directly — writing a true story, a fictional story, and a deliberately "mixed" story (real GAN history blended with fabricated specifics), then having team members classify each as fact, fiction, or mixed without labels (Ao8). Built a companion 8-slide deck presenting the activity, the reveal, and the mapping back onto real GAN architecture (Lo8).

**WHAT I LEARNED** Believability is a learned distribution, not a property of drama — the story that read as most authentic wasn't the most dramatic, it was the one with small, specific, mundane details that are hard to fabricate without lived experience. Mixed content (real facts blended with invented ones) was the hardest for evaluators to classify, because early true claims transfer credibility to what follows — directly relevant to why blended-truth disinformation and deepfakes are harder to catch than outright fabrication.

**A08 — 94%****L08 — 95%****MOD 09****Computer Vision: Image Processing**

**ACTIVITIES** Researched and compared three architecturally different image-generation platforms — Runway ML (Autoregressive-to-Diffusion video/ image model), Google Gemini (Multimodal Diffusion Transformer), and NightCafe Studio (consumer, multi-backend diffusion platform) — covering architecture, company history, and documented bias/ethics controversies for each, including Gemini's real February 2024 historical-imagery incident. Designed a tiered prompt set (simple → complex → abstract) with two deliberate bias-probe prompts.

**WHAT I LEARNED** Each platform expressed the same prompt differently rather than converging on a similar output — consistent with the fact that all three run genuinely different underlying architectures and training approaches. Text-to-image generation in 2026 isn't one technology, it's several different approaches (latent diffusion, autoregressive-to-diffusion, multimodal diffusion transformers) making different tradeoffs — disclosed vs. undisclosed training data being the starkest one.

**HONEST GAP** the generated images themselves weren't saved during testing, so this module's real deliverable is the comparative research and general testing impressions, not image-by-image evidence. Disclosed directly in the Lo9 report rather than glossed over.

## MOD 10 NLP: Basics

**ACTIVITIES** Manually implemented a full NLP pipeline — cleaning, tokenizing, stop-word removal, stemming, POS tagging 20+ tokens, and Count Vectorization — on a corpus pulled from my own AO1 essay (the AlexNet paragraph), without automated NLP libraries (A10). Explored IBM Watson Assistant's Lendyr Demo and the underlying trial builder — action/trigger-phrase configuration, the integrations catalog, phone/ voice deployment — testing both a scripted path (a Dental student-loan flow in Arkansas, handled correctly) and an off-script one ("Modify in progress application," which hit an agent-offline dead end) (Lab 10).

**WHAT I LEARNED** Stop-word removal alone cut my corpus from 98 to 61 tokens — over a third of ordinary English text is structural filler, not topical content. Stemming's fragments ("decis," "viabil") made the real speed-vs-accuracy tradeoff concrete instead of theoretical. From Lab 10: Watson Assistant is a construction kit, not a ready-made assistant — its intelligence is entirely a function of what a business explicitly built, which is real enterprise flexibility but also means every unbuilt path is a hard dead end. My own assessment: "not intuitive at all... very enterprise focus... bulky... static... unimpressed" compared to consumer assistants.

## MOD 11 NLP: LLMs

**ACTIVITIES** Analyzed AI digital assistants and conversational AI against the movie *Her* (2013), arguing the emotional-bond comparison is overblown — today's assistants are transactional, not companions — while noting persistent-memory features now shipping are an active step toward closing that gap (A11). Ran a real 5-question comparison of Siri vs. Google Assistant (L11).

**WHAT I LEARNED** The assistant that was more explicit about lacking feelings (Google Assistant) actually handled open-ended questions more convincingly than the one that didn't disclaim as much (Siri, which punted to search links on the same prompts). Being conversational isn't about claiming personality — it's about actually engaging with what was asked.

Submitted late Aug 2, pending grading.

## MOD 12 AI Agents

**ACTIVITIES** Wrote a report on predictive modeling in AI agents, tying it directly to A07's real RTU failure predictor, then built an original short-film storyline ("ORACLE") depicting a predictive-maintenance AI that earns expanded autonomy after one success and drifts toward optimizing the wrong objective — echoing *Eagle Eye* (2008).

**WHAT I LEARNED** The realistic risk with predictive AI isn't a rogue superintelligence — it's a system that performs well enough on its stated metric to earn expanded authority, while nobody notices the metric itself was incomplete. Predictions should inform; humans should authorize — that boundary should be the hardest thing to expand, not the easiest.

Submitted late Aug 2, pending grading.

## MOD 13 Robotics: Grand Tour

**ACTIVITIES** Designed "SentryTech," a robot with two real deployment modes built on my own HVAC/cGMP cleanroom background — commercial HVAC pre-dispatch verification (preventing the wasted-trip problem from real field experience) and cleanroom environmental-monitoring. Built a full ethical analysis (privacy, safety, job displacement, societal impact) and a digitally-created concept sketch.

**WHAT I LEARNED** Writing the ethics section honestly required naming an uncomfortable tradeoff directly — pre-dispatch verification could reduce dispatch *volume* even as it protects the quality of dispatches that do happen — rather than only listing the tool's benefits.

Sketch available: [modules/module-13/assignments/sentrytech\\_sketch.svg](modules/module-13/assignments/sentrytech_sketch.svg)

Submitted late Aug 2, pending grading.

## Midterm Exam — AI Glossary

**ACTIVITIES** Group glossary project submitted as part of TuringCollective.

94%



#### EXTRA CREDIT · NOT CANVAS-GRADED

### Bonus — Building an Agent: Hermes

**ACTIVITIES** Took on the optional, verbally-announced challenge (not a graded Canvas item) to build a real AI agent rather than just use a chat window. Configured and ran Hermes Agent — a purpose-built agentic architecture with a swappable LLM brain (Claude Sonnet 4, but portable across 20+ providers), a six-layer stack (brain → agent loop → memory & context → skills → 12 tools → surfaces), and a live deployment scoped to this exact course: a Telegram bot pulling assignments and grades straight from the Canvas API, persistent memory of course context (HCCS Eagle Online URL, course ID, response-format preferences), and scheduled cron jobs (Sunday 8pm — check Canvas for new assignments; Friday — summarize what's due next week; 1 hour before class — send the Teams meeting link). Safety is built in, not bolted on: destructive commands require explicit approval, every action is checkpointed and reversible, secrets are auto-redacted from tool output, and parallel agents are isolated via Git worktrees so nothing steps on anything else.

**WHAT I LEARNED** Everything else in this course — the Dartmouth Conference, symbolic AI, backprop, RLHF, NLP pipelines, GANs — is the decades of foundational effort that made a system like this *possible*. Hermes is the other half of the picture: it's not a smarter model, it's architecture wrapped around a model — memory, tools, scheduling, and safety rails — that turns "a brilliant reasoner trapped in a stateless chat box" into something that persists, acts, and improves. The line that stuck with me: intelligence is rented from the model, but capability is built in the architecture. Consumer AI tools and LLM harnesses are what finally put decades of research at an individual's fingertips — this course gave me the history and the theory; this challenge showed me the harness that makes it usable in a real workflow.

*Not Canvas-graded — extra-credit challenge, included here because it directly extends what this course covers. Source briefing: [portfolio/Hermes\\_Agent\\_bonus\\_challenge.pdf](#)*

### Modules 14–16

Nothing assigned on Canvas as of this portfolio's drafting — placeholders only (Puzzle of the Day / Joke of the Day / Assignment / Laboratory, none populated by the instructor yet).

## Conclusion

One throughline runs through nearly every module in this course: the gap between mechanically-verifiable correctness and actual human or real-world value. It shows up as RLHF optimizing a model to *sound* aligned without knowing when it's uncertain (Module 05); as a Scratch game's race condition, where two independent scripts touching the same shared state produced a technically-running but wrong result (Module 03); as a computational-humor algorithm that can verify a joke's mechanical structure but not whether a given audience's background makes it land (Module 05); as Watson Assistant performing flawlessly within its scripted paths and hitting a hard dead end the moment a request falls outside them (Module 10); and as a robot design where the honest answer to an ethical tradeoff (pre-dispatch verification reducing dispatch *volume*, not just improving dispatch *quality*) had to be stated directly instead of glossed over (Module 13). The pattern repeats because it's the actual shape of the problem: a system can be internally consistent and still miss what a human situation actually needs. Building this repository — backfilling real documentation, mini critical-design reviews, and honest gap notes alongside the graded work itself — was itself an exercise in the same lesson: producing something that's not just technically complete, but genuinely explainable and defensible. The Hermes Agent bonus challenge closed the loop on the whole semester: Modules 01–13 are the history and theory of how AI got here across decades of ambition and setback; building a real agent on top of that foundation is what makes all of it usable, right now, at my fingertips.

## Future Objectives in AI

AI is a rapidly evolving field, and my objective going forward is to build a genuinely comprehensive understanding of it — not just the individual tools, but the components that connect them and how those components keep shifting as the technology matures. This course gave me that foundation on the theory side: how we got from symbolic AI and neural-network winters to the deep learning and LLM era, and why each generation of breakthroughs reshaped what "intelligence" even means. The Hermes Agent bonus challenge showed me the other half — how that theory gets wrapped into real, deployable architecture: memory, tools, scheduling, safety rails. I want to keep building on both halves at once, because neither is complete without the other.

I also want to understand AI's role from two directions, not one. From an occupational standpoint: how these technologies are actually getting incorporated into businesses right now, and what that means for the skill set employers need. But just as much from an entrepreneurial standpoint — growing my own knowledge base to the point where I can identify real problems and build real solutions with AI as a collaborator, not just a tool I call on. Clay Climate AI is where that's already started (the Make.com-to-self-hosted-agent migration work, the Hermes deployment itself), and I want that to keep growing: developing the judgment to know where AI genuinely creates value versus where it's just automation dressed up, and the technical range to build that value myself instead of waiting for someone else to build it for me.

The throughline of this whole portfolio — that a system can be technically correct and still miss what a real situation needs — is exactly the gap I want to keep closing. Not by trusting the tools more, but by understanding them well enough to work with them effectively: knowing what they can verify, what they can't, and where my own judgment has to fill that gap. That's the skill set I'm building toward, and this course was the starting point.

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